Gustabo Lazaro

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Engineering Portfolio: https://glazaro3251-dev.github.io/Gustabo_Lazaro_ME/

EDUCATION

California State Polytechnic University, Pomona (CPP) — ***Expected August 2026***

Bachelor of Science, Mechanical Engineering

Related Coursework: Manufacturing Processes, Mechanical Design, Engineering Materials, Measurements and Instrumentation, Finite Element Analysis (FEA), Control of Mechanical Systems, Engineering Analysis, Thermodynamics, Fluid Mechanics

WORK EXPERIENCE

United Rentals Inc. - Fontana, CA — ***June 2011 — Present***

Service Technician III

• Inspect, troubleshoot, and repair mechanical, electrical, and hydraulic equipment/ tools to keep units safe, reliable, and ready for use.
• Use service manuals, schematics, inspection results, and equipment symptoms to identify problems and determine the proper repair approach.
• Document repair findings, parts used, completed work, and service recommendations to support accurate records and future troubleshooting.
• Work with service team members, parts personnel, management, and customers to prioritize repairs, resolve equipment issues, and return units to service.

STUDENT INVOLVEMENT

CPP NGCP (Northrop Grumman Collaboration Project) - Pomona, CA — ***May 2025 — Present***

Payload Integration Member

- Overseeing cross-functional coordination with subsystems leads to track progress, resolving technical challenges, and maintain project schedules.
- Co-led a team in the design, prototyping, and testing of UAV payload systems for aerial deployment.
- Built a MATLAB model for 3D parachute trajectory simulation to enhance drop accuracy and reliability.
- Designed and implemented a data logging system using Arduino, accelerometer, and SD card module to monitor and verify payload acceleration below 2Gs during deployment.
- Developed and tested autonomous payload release mechanisms in SolidWorks for a UAV supporting relief package delivery missions.
- Designed a payload enclosure with a parachute detachment mechanism by collaborating with Unmanned ground vehicles (UGV) to improve efficiency of picking up the medical relief package.
- Tested various parachute folding configurations and analyzed impact forces using energy-based methods; validated results with MATLAB Monte Carlo simulations.
- Conducted FEA and CFD analysis on the payload system to assess structural integrity and aerodynamic behavior during descent.
- Deployed the payload within 250ft of the specified release envelope, meeting overall project goals.

Finite Element Analysis Project — FEMAP/Nastran

• Prepared a simplified finite element model by removing small non-structural components and organizing major parts for analysis.
• Assigned material properties, shell thicknesses, mesh types, and joint connections to support structural analysis setup.
• Documented modeling assumptions, constraints, and analysis limitations as part of an individual and team-based FEA project.

Skills

CAD & Design: SolidWorks, AutoCAD 2D

Simulation & Analysis: MATLAB/Simulink, NX NASTRAN (FEMAP), SolidWorks Simulation

Other: 3D printing, Arduino IDE, Microsoft Office (Excel, Word, PowerPoint), Hands-On experience with hand tools and power tools, MIG Welding